ACS APPLIED ENERGY MATERIALS

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Thermoelectric Generator Using Camphorsulfonic Acid-Doped Copper Benzene-1,3,5-Tricarboxylate Metal−Organic Frameworks with Polyaniline Coating

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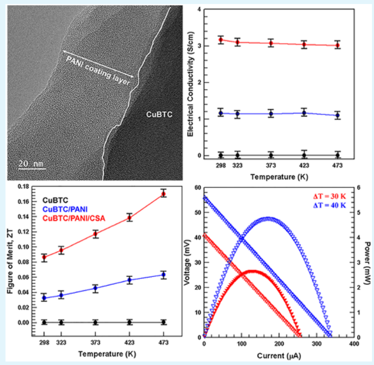
Cite This: ACS Appl. Energy Mater. 2025, 8, 7363−7370 Read Online



ABSTRACT: Metal−organic frameworks (MOFs), which comprise metal cations and organic ligands connected by coordination bonds, exhibit exceptional porosity and tunable properties, making them promising materials for thermoelectric (TE) applications. However, most MOFs have low electrical conductivity owing to limited conjugation and poor orbital overlap between ligands and metal ions. To address this challenge, herein, we designed and evaluated MOF-based hybrids with customizable thermal and electrical properties. By integrating conjugated polymers with MOFs, high conductivity for TE generation can be achieved. Copper benzene-1,3,5-tricarboxylate (CuBTC) was selected because of its high Seebeck coefficient at near room temperature. First, to increase the low electrical conductivity of CuBTC, it was coated with highly conductive (360 S/cm) polyaniline (PANI) (CuBTC/PANI). Then, camphorsulfonic acid (CSA) was used to dope CuBTC/PANI and further enhance its conductivity. Consequently, CuBTC/PANI/CSA is 1000 times more conductive than pristine CuBTC. A TE generator using CuBTC/PANI/CSA exhibited excellent performance with an output power density of 23.8 mW/cm^2 at a temperature difference of 40 K. This significant improvement in TE performance enables the utilization of MOF-based hybrids in advanced TE applications, particularly for wearable and flexible devices.

KEYWORDS: *metal−organic framework, conducting polymer, polyaniline, Seebeck coefficient, thermoelectric generator*

1. INTRODUCTION

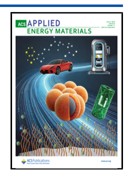
Thermoelectric (TE) materials directly transform thermal energy to electrical energy. This unique property makes them useful for addressing the energy requirements of flexible and wearable devices.[1−4] Figure of merit ZT determines the efficiency of TE materials and is defined as

$$\mathrm{ZT} = \frac{S^2 \cdot \sigma}{\kappa} \cdot T$$

where S, σ, κ, and T represent the Seebeck coefficient, electrical conductivity, thermal conductivity, and absolute temperature, respectively. The ZT value, and consequently, the TE performance, can be improved by boosting S or σ or by decreasing κ. Previously, several methods for enhancing the performance of TE materials have been investigated, such as band structure optimization,[5,6] introducing point defects by elemental doping or adding vacancies,[7,8] scaling the material to the nanoscale to promote more phonon scattering at the interfaces,[9,10] and incorporating heteromaterials to improve the power factor or decrease thermal conductivity.[11,12]

Metal−organic frameworks (MOFs) are porous crystalline compounds consisting of an organic ligand interconnected by a coordination bond with a metal cation.[13−16] MOFs have exceptional porosity and customizable composition and structure, making them highly promising TE materials. However, most MOFs act as electrical insulators due to the limited conjugation and poor orbital overlap between organic ligands and metal ions.[17,18] Fortunately, by carefully selecting and arranging the metal centers and organic ligands, the thermal and electrical properties of MOFs can be precisely tuned. Because MOFs are typically synthesized in a powder form, processing them into solid supports for wearable design applications is a critical challenge. MOF-based hybrid materials with customizable thermal and electrical properties have been proposed to further optimize their TE performance. Xue et al. boosted the TE performance of zeolitic imidazolate frameworks (ZIF-67) via in situ growth on carbon nanotubes.[19] By integrating MOFs with materials with good electrical properties, composites with excellent TE performance can be achieved. In particular, integrating conjugated polymers with

Received: March 12, 2025
Revised: April 24, 2025
Accepted: May 20, 2025
Published: May 26, 2025



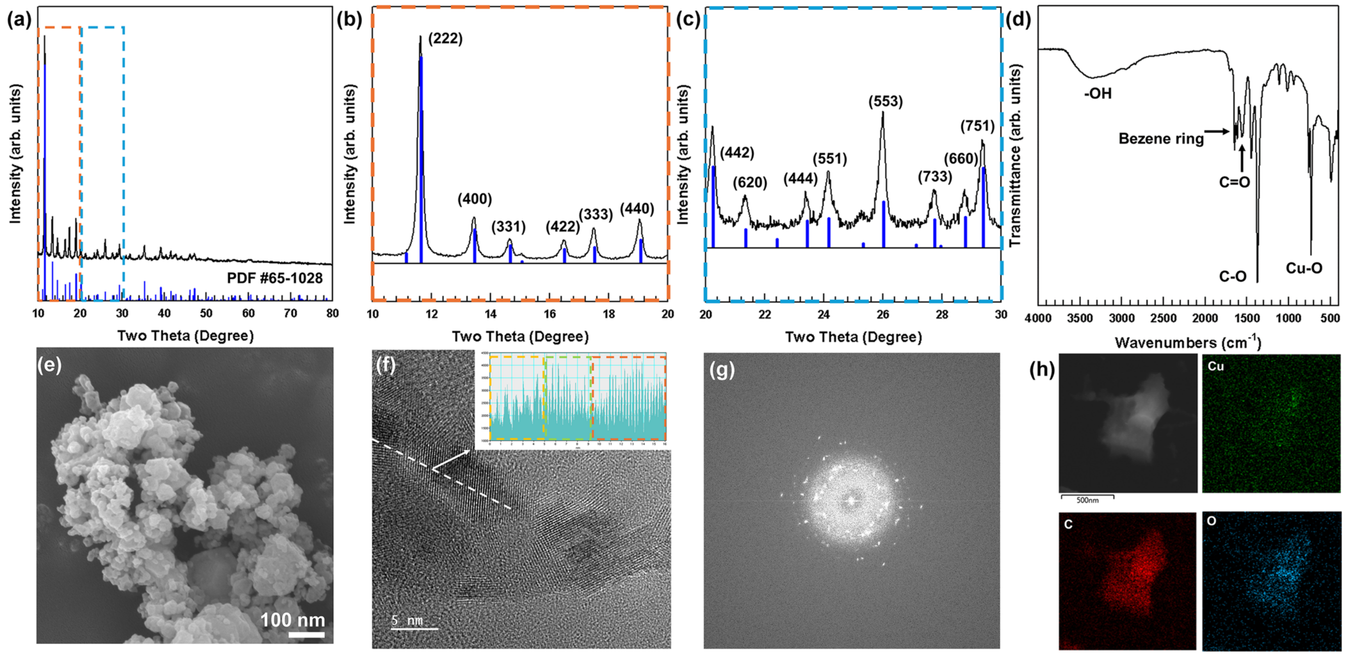


Figure 1. (a) XRD and (b, c) high-resolution XRD patterns, (d) FTIR spectra, (e) FE-SEM image, (f) FE-TEM image, (g) selected-area electron diffraction pattern, and (h) EDS elemental mapping profile of CuBTC.

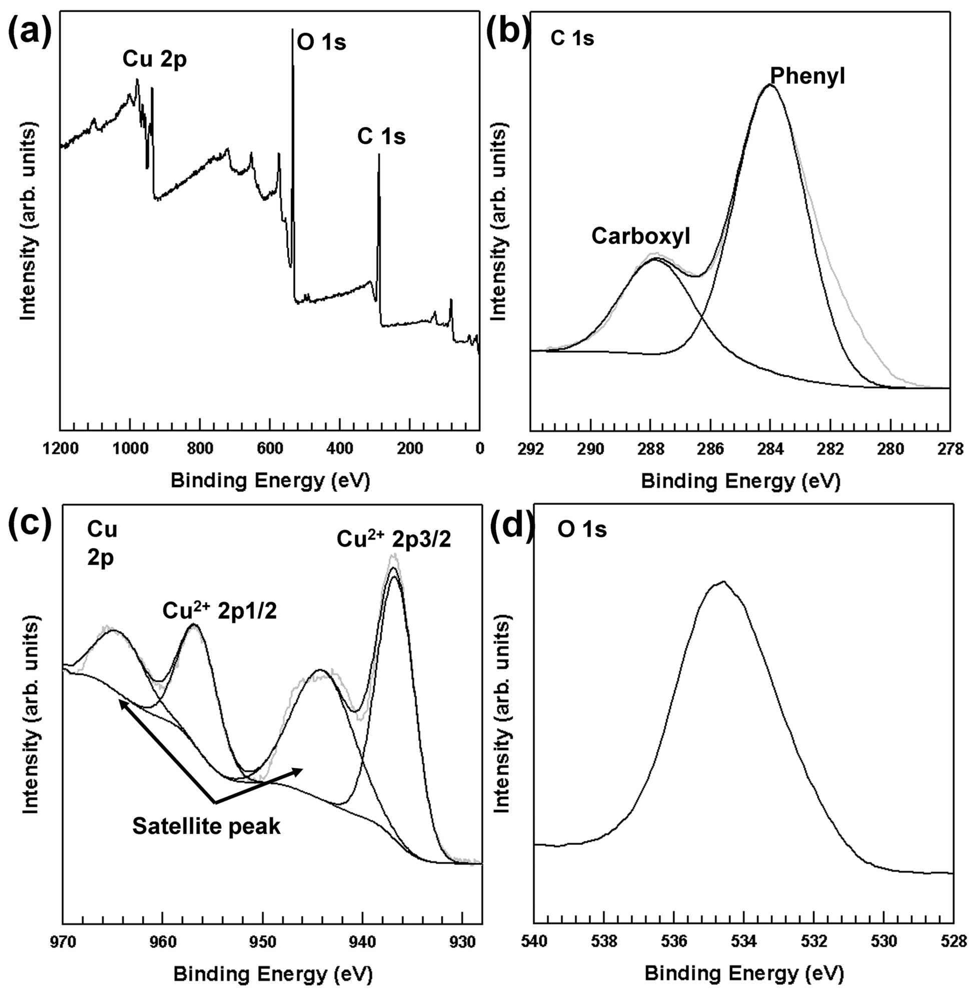


Figure 2. (a) XPS survey spectrum and high-resolution XP spectra of (b) C 1s deconvolution (c) Cu 2p deconvolution (d) O 1s of CuBTC.

https://doi.org/10.1021/acsaem.5c00727
ACS Appl. Energy Mater. 2025, 8, 7363−7370

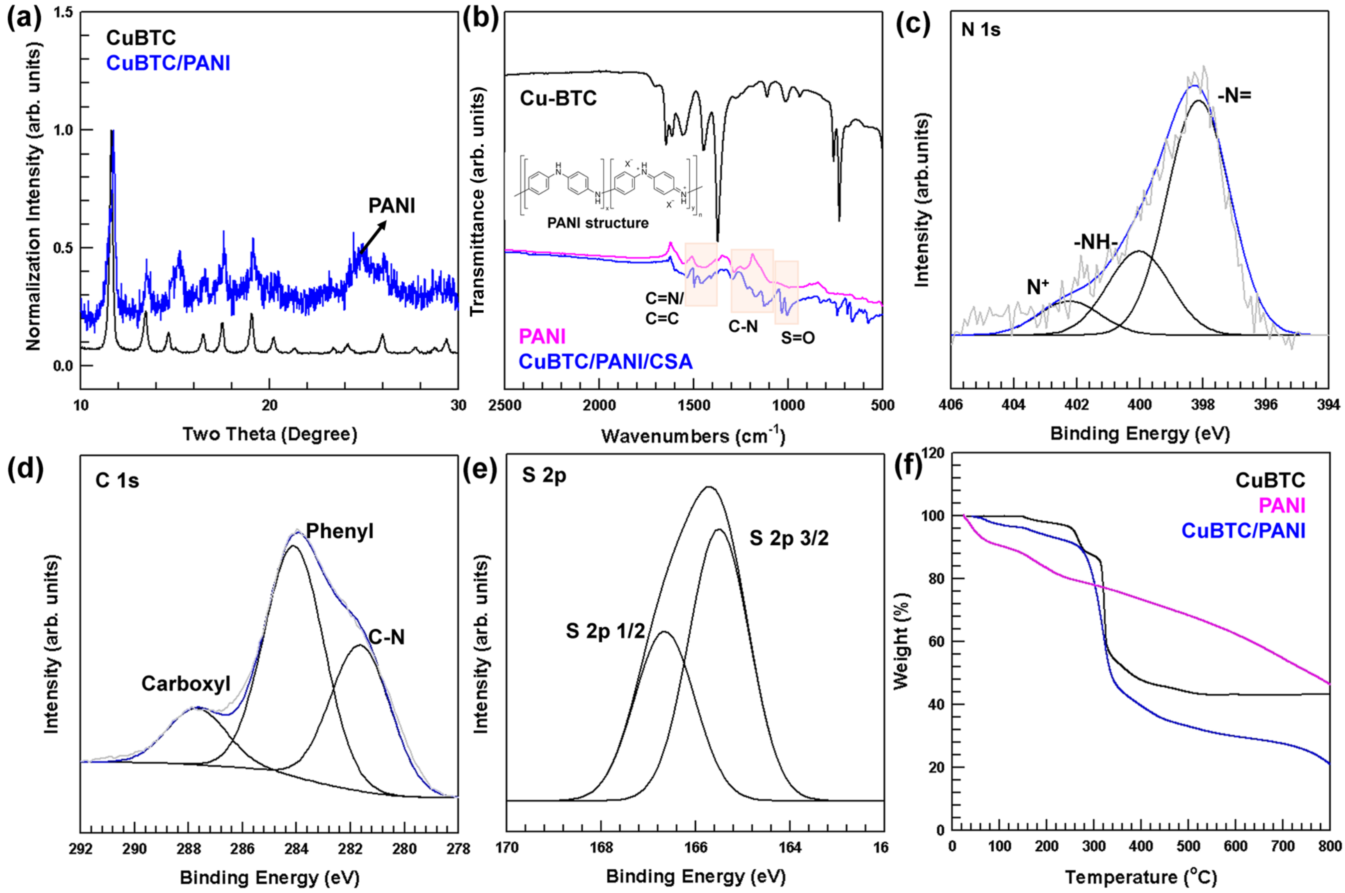


Figure 3. (a) XRD pattern and (b) FTIR spectra of CuBTC/PANI/CSA. High-resolution XP spectra of (c) N 1s deconvolution, (d) C 1s deconvolution, and (e) S 2p deconvolution of CuBTC/PANI/CSA. (f) TGA results of CuBTC and CuBTC/PANI.

metal−organic coordination polymers can increase the electrical and thermal conductivities of MOFs.

Copper benzene-1,3,5-tricarboxylate (CuBTC) has a high Seebeck coefficient at near room temperature (∼400 μV/K at room temperature),[20] but extremely low electrical conductivity.[20,21] Joseph et al. increased the electrical conductivity of CuBTC using poly(3,4-ethylenedioxythiophene) polystyrenesulfonate (PEDOT:PSS).[19] Meng et al. conducted the in situ reduction of core−shell-structured CuTCNQ@CuBTC to improve the electrical conductivity of CuBTC.[22]

In this work, CuBTC was coated with highly conductive polyaniline (PANI) (360 S/cm) to form CuBTC/PANI, which exhibits enhanced conductivity in relation to that of pristine CuBTC. Next, CuBTC/PANI was doped with camphorsulfonic acid (CSA) to further increase the electrical conductivity. Previous studies reported the ability of CSA to increase the conductivity of PANI.[23−25] Subsequently, the TE properties of CuBTC/PANI/CSA were characterized, and a TE generator (TEG) using CuBTC/PANI/CSA was fabricated and its TE performance was evaluated.

2. RESULTS AND DISCUSSIONS

2.1. Confirmation of CuBTC MOF Synthesis. CuBTC particles were synthesized following a protocol established in previous studies.[21] A detailed description of the synthesis and characterization of the materials is provided in the Supporting Information. Figure 1(a−c) presents the X-ray diffraction (XRD) patterns, confirming the successful construction of CuBTC; these results are consistent with previously reported data (PDF #65−1028). The lattice parameters can be estimated from the XRD results.[26] The calculated lattice parameter is 26.2764 ± 0.007 Å, which aligns with the specifications in PDF #65−1028. Fourier transform infrared (FTIR) spectroscopy was also performed. As revealed in Figure 1(d), the FTIR peak at 700 cm^{-1} is assigned to the Cu−O bond. The FTIR peaks at 1300, 1700, and 3400 cm^{-1} are assigned to the C−O, C═O, and −OH bonds of BTC, respectively. The complex peaks at 1500−1600 cm^{-1} indicate benzene rings from benzene-1,3,5-tricarboxylic acid.[27] The CuBTC particle framework and particle size were analyzed through field emission scanning electron microscopy (FE-SEM) and field emission transmission electron microscopy (FE-TEM). The FE-SEM images in Figure 1(e) show the CuBTC particles with varying size distributions. Furthermore, in Figure 1(f), the FE-TEM images express the polycrystalline structure of CuBTC. Three types of lattice fringes can be seen from the white dashed line of the high-resolution FE-TEM image, and the selected-area electron diffraction pattern in Figure 1(g) verifies the polycrystalline structure of the CuBTC particles. Energy dispersive X-ray spectroscopy (EDS) elemental mapping reveals uniform distribution of C and O. Cu is present not only in the particles but also on the entire surface, owing to the copper frame of the FE-TEM grid.

The XPS and high-resolution Cu 2p and C 1s spectra of CuBTC are shown in Figure 2. The XPS survey spectrum reveals the existence of C, O, and Cu in CuBTC samples. The high-resolution C 1s spectra indicates the presence of carboxyl and benzene rings in the BTC structure. The Cu 2p spectra

https://doi.org/10.1021/acsaem.5c00727
ACS Appl. Energy Mater. 2025, 8, 7363−7370

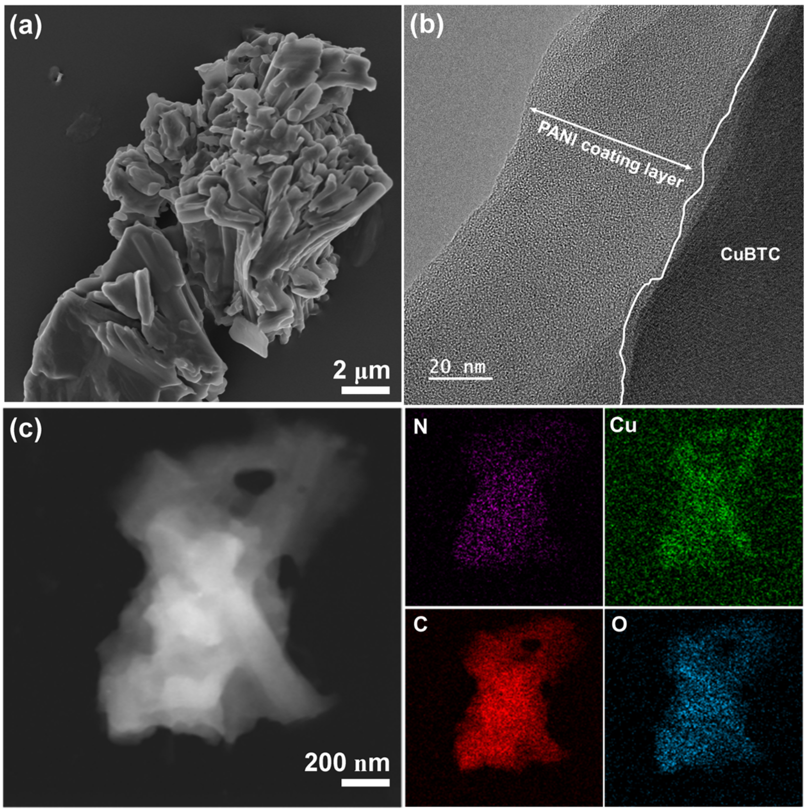


Figure 4. (a) FE-SEM and (b) FE-TEM images and (c) EDS elemental mapping of CuBTC/PANI.

confirm that the ionic state of Cu in CuBTC was Cu^{2+} with satellite peaks.

2.2. PANI Coating of CuBTC. The XRD pattern of CuBTC/PANI is presented in Figure 3(a), which aligns with that of pristine CuBTC. The crystal structure of CuBTC is monoclinic with space group $P21/n$ and the lattice parameters are a = 6.7654, b = 18.8135, c = 8.5144; $\alpha = \gamma = 90°$, β = 92.439°.[28] The PANI coating process does not affect the crystal structure of CuBTC, and the characteristic XRD pattern of PANI appears at 25°.[29] Figure 3(b) displays the FTIR spectrum of undoped PANI. The corresponding FTIR peak of C═C stretching vibrations of the quinoid and benzenoid rings occur at 1581 and 1494 cm^{-1}, respectively. The FTIR peaks at 1303, 1244, and 1150 cm^{-1} also indicate the signature peaks of benzenoid quinoid rings.[24] The FTIR and XRD analyses both confirm the presence of undoped PANI, which is characterized by an emeraldine structure containing a benzenoid ring and quinoid rings. The FTIR spectrum of CuBTC/PANI in Figure 3(b) reveals peaks at 1591, 1490, 1300, 1240, and 1151 cm^{-1}, which are attributed to PANI covering the CuBTC surface.[23] The weakening of the CuBTC FTIR spectrum indicates that the complete coating of the CuBTC surface by PANI, as further verified by the FE-SEM images (Figure 4(a)). Additionally, the FE-TEM image in Figure 4(b) shows a 40−50 nm-thick PANI layer on the CuBTC surface. Further characterization by XPS reveals spectra containing the N 1s and C 1s peaks of CuBTC/PANI/CSA, as shown in Figure 3(c−e). The peaks at 398.15, 400.04, and 402.25 eV correspond to the previously reported emeraldine-based-PANI-related peaks. The binding energy of 281.71 eV is associated with the carbon bond of emeraldine-based PANI and the binding energies of 284.01 and 287.74 eV are associated with the carbon bond of CuBTC (as shown in Figure 2(b)).[30] Thermogravimetric analysis (TGA) was performed to determine the amount of coated PANI. The weight of CuBTC decreased because of the presence of O atoms in BTC. After coating with PANI, thermal decomposition starts at approximately 300 °C, indicating the decomposition of PANI. A comparison of the TGA results of CuBTC before and after PANI coating confirms that CuBTC/PANI can withstand high temperatures required for the analysis of TE performance. In Figure 4(c), EDS elemental mapping shows a uniform dispersion of N on the surface of CuBTC/PANI particles. Similar to the EDS elemental mapping results of CuBTC (Figure 1(h)), Cu is present not only in the particles but also on the entire surface due to the copper frame of the FE-TEM grid.

2.3. TE Properties of CuBTC/PANI. Parameters S and σ were measured to verify the effect of the PANI coating on TE performance. Pristine CuBTC exhibited an extremely low σ of 5×10^{-3} S/cm and a high S of 401.16 μV/K at room temperature. This low σ limits the application of CuBTC in TE modules. Therefore, conducting polymer PANI was used to coat CuBTC and improve its σ. After PANI coating, σ increased to 1.16 S/cm, which is 1000 times higher than that of pristine CuBTC. CSA doping further enhanced the σ of CuBTC/PANI. This σ enhancement was confirmed by measuring the carrier concentration (n) and mobility (μ) of the composites. After PANI coating and CSA doping, n and μ increased. Parameter n increased from 7.99×10^{15} to 7.96×10^{17} and 1.05×10^{18} cm^{-3}, whereas μ increased from 4.03 to 9.11 and 18.87 $cm^2/(V\ s)$. These changes in n and μ indicate improvement in σ and a slight decrease in S, which is inversely proportional to $n^{2/3}$.[31] Carrier mobility can be improved by

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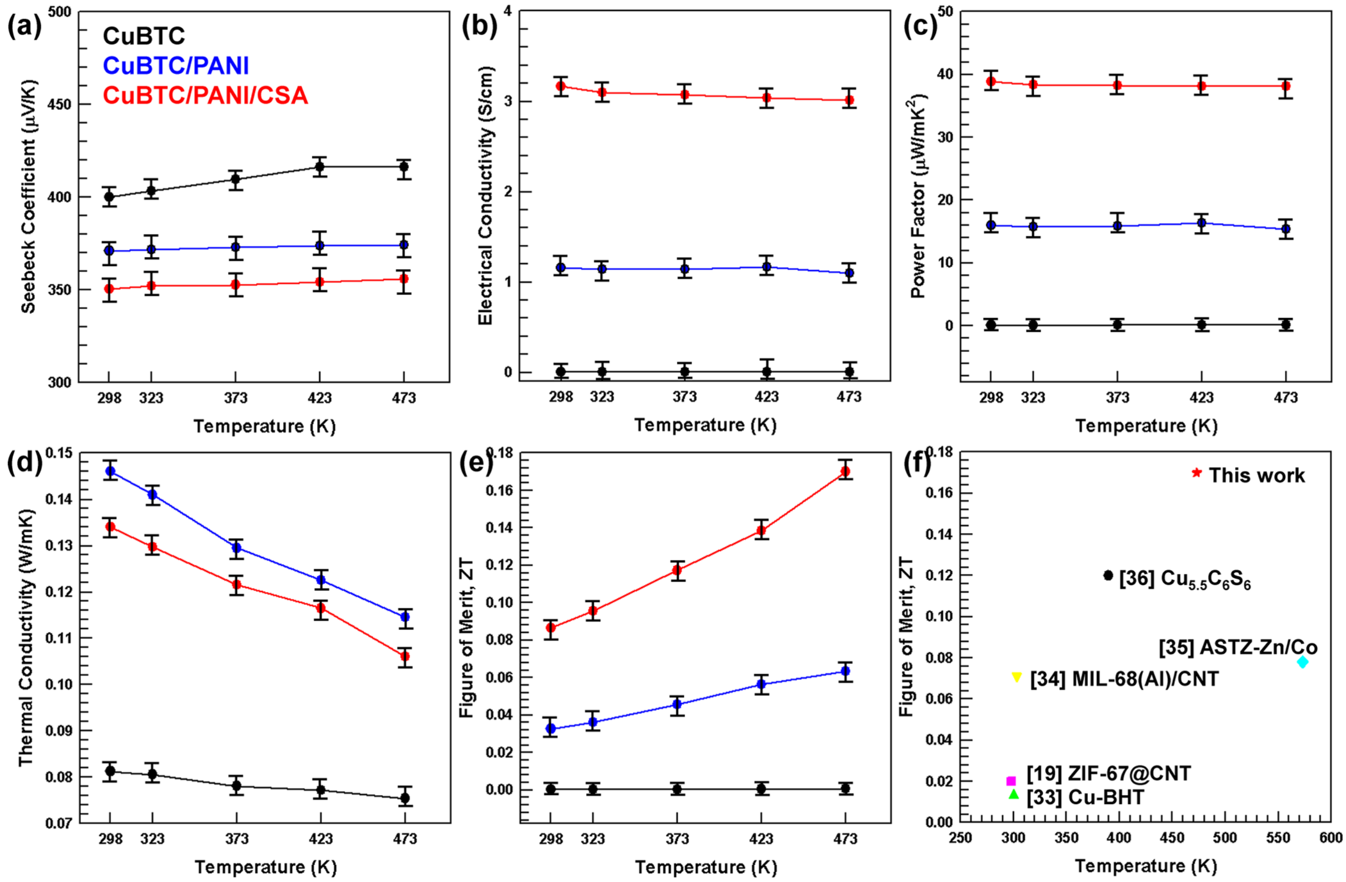

Figure 5. (a) Seebeck coefficient, (b) electrical conductivity, (c) calculated PF, (d) thermal conductivity, and (e) ZT value of the CuBTC, CuBTC/PANI, and CuBTC/PANI/CSA composites with increasing temperature. (f) ZT values of different MOFs.[19,33−36].

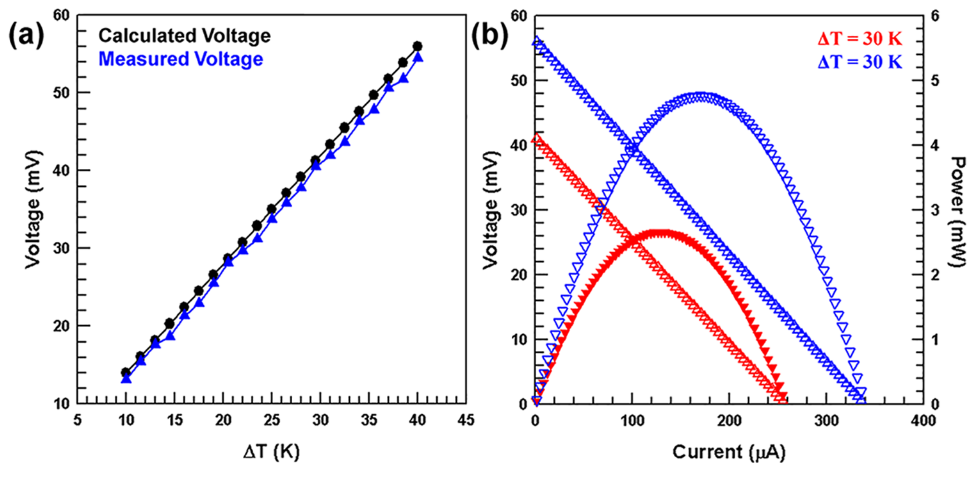

Figure 6. (a) Open circuit voltage and (b) output power of the fabricated TEG using four CuBTC/PANI/CSA legs.

$\pi-\pi$ interactions with the benzene ring of the coated PANI structure. For the CuBTC/PANI and CuBTC/PANI/CSA composites, with increasing temperature, S exhibits nearly stable values of 373 and 352 μV/K, respectively, whereas the corresponding σ values decrease slightly. Power factor (PF) calculations show that CuBTC/PANI/CSA exhibits the highest PF value of 38.8 μW/mK^2 at room temperature. However, from room temperature to 473 K, the PF remains almost constant because the change in S and σ with temperature is not significant. The thermal conductivities of the composites were measured; they were found to gradually decrease with increasing temperature. The total thermal conductivity consists of both lattice (κ_L) and electrical (κ_e) components. The electrical thermal conductivity, κ_e, can be approximated using the equations, as shown below

$$\kappa = \kappa_L + \kappa_e$$

$$\kappa_e = L \times T \times \sigma$$

$$L = \left[1.5 + \exp\left(-\frac{|S|}{116}\right)\right] \times 10^{-8}\ \mathrm{W{\cdot}\Omega{\cdot}K^{-2}}$$

https://doi.org/10.1021/acsaem.5c00727
ACS Appl. Energy Mater. 2025, 8, 7363−7370

where L denotes the Lorentz number.[32] The electrical thermal conductivity was determined using the calculated Lorentz number and the measured electrical conductivity. The value of κ_L was calculated by subtracting κ_e from the total thermal conductivity. Importantly, κ_L is the primary contributor to the overall thermal conductivity of the composites, given the very low value of L, which results in a significantly lower κ_e. The calculated κ_e value is ∼2.2×10^{-3} W/mK for CuBTC/PANI/CSA at 473 K, which is the highest among the composites. This indicates that the contribution of the electrical thermal conductivity is extremely lower than the lattice thermal conductivity. κ_L can be decreased by phonon scattering at the heterointerfaces of the composites. With increasing temperature, phonons can be easily scattered at the interfaces, which decreases thermal conductivity. Reduced κ and enhanced PF resulted in the peak ZT value of 0.17 at 473 K. Figure 5(f) compares the TE properties of CuBTC/PANI/CSA with other recently studied MOFs below 600 K. The results of this study confirm that PANI coating followed by CSA doping drastically improves the TE performance of CuBTC.

2.4. TEG Using CuBTC/PANI/CSA.

A TEG was fabricated comprising CuBTC/PANI/CSA legs on a polyimide substrate. Low resistance Cu wires and Ag pastes were used to attach the CuBTC/PANI/CSA legs. The TE legs were rectangular with dimensions of 0.1 cm × 0.5 cm. Figure 6 shows the voltage (V) and power (P) output graphs of the TEG. The TEG voltage output ranges from 13 to 54 mV for temperature differences (ΔT) between 10 and 40 K. It is noteworthy that V has a linear correlation with the temperature difference. V can be calculated by the following formula

$$V = |S| \cdot \Delta T \cdot N$$

where N is the number of TEG legs, which is four in this study. Observed V is slightly lower than the theoretical value. This is because the theoretical calculation does not account for the resistances of the Cu wire and Ag paste. Output power P can be determined using the following equation

$$P = I^2 \cdot R_{load} = \left(\frac{V_{oc}}{R_{in} + R_{load}}\right)^2 \cdot R_{load}$$

where I is the output current, R_{load} is the variable load resistance connected in parallel, and R_{in} is the resistance of the TEG. Additionally, R_{in} is a constant that depends on the material and temperature.[24] Because R_{in} is determined by the number of TEG legs, the change in R_{load} is used to determine peak P. The TEG achieves its maximum P when R_{load} equals R_{in}, resulting in a distinct parabolic output. The maximum P obtained is 4.76 mW at a temperature difference ΔT = 40 K, with a power density of 23.8 mW/cm^2. Our results verify that the CuBTC/PANI/CSA TEG can be used as an energy source for small devices typical in wearable electronics.

3. CONCLUSIONS

In this study, we successfully synthesized CuBTC MOFs and coated them with PANI to enhance their TE properties. The synthesized CuBTC particles were characterized through XRD, FTIR spectroscopy, and electron microscopy, which collectively validated the structure and morphology of the material. The PANI coating was characterized by FTIR spectroscopy, XRD, FE-SEM, FE-TEM, and XPS, confirming the uniform and successful coating of the CuBTC particles by PANI.

The TE performance of the CuBTC/PANI composites significantly surpassed that of pristine CuBTC. Coating with PANI increased the electrical conductivity of CuBTC by 3 orders of magnitude, while maintaining a high Seebeck coefficient. This enhancement was attributed to the increased carrier concentration and mobility caused by the $\pi-\pi$ interactions of the benzene rings in PANI. Furthermore, the thermal conductivity decreased with an increase in the temperature, mainly due to the reduction of lattice thermal conductivity due to phonon scattering at the interface. The CuBTC/PANI composite exhibited a maximum TE efficiency of 0.17 at 473 K, indicating promising TE performance.

A TEG was fabricated using the CuBTC/PANI/CSA composite, and its ability to harvest energy was successfully demonstrated. The TEG exhibited a maximum output power of 4.76 mW and power density of 23.8 mW/cm^2 at a temperature difference of 40 K, highlighting its potential for use in wearable devices and other low-power applications.

The CuBTC/PANI composites synthesized, characterized, and applied in this study promote the significant advancement in the development of MOF-based TE materials. The combination of enhanced electrical conductivity and high Seebeck coefficient ensures high TE performance. This study opens new avenues for the application of MOF-based composites in TE devices and energy-harvesting technologies. Future research should focus on optimizing the PANI coating process, further reducing the thermal conductivity and exploring the scalability of these materials for commercial applications.

■ ASSOCIATED CONTENT

Data Availability Statement

Data will be made available on request.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsaem.5c00727.

Experimental section including materials; synthesis of CuBTC and PANI coating; fabrication of CuBTC/PANI/CSA TEG and characterization; crystal structure of CuBTC (PDF)

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Notes

The authors declare no competing financial interest.

https://doi.org/10.1021/acsaem.5c00727
ACS Appl. Energy Mater. 2025, 8, 7363−7370

ACKNOWLEDGMENTS

This work was supported by the Human Resources Development (No.20214000000280) of the Korea Institute of Energy Technology Evaluation and Planning (KETEP) grant funded by the Korean government Ministry of Trade, Industry, and Energy.

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https://doi.org/10.1021/acsaem.5c00727
ACS Appl. Energy Mater. 2025, 8, 7363−7370